

Finals Practice Task 01. Data Prep And EDA using SQL

Description:

The activity focused on utilizing MySQL Workbench to clean and organize the dirty_shipments dataset. This process involved addressing various data issues such as inconsistent formatting, missing information, negative values, and incorrect entries to ensure the dataset's reliability. To enhance the dataset's usefulness, additional columns were created to calculate shipment and delivery time differences. This step emphasized the significance of maintaining data accuracy, which is crucial for analyzing shipment performance and making informed logistical decisions. The overall goal was to improve data quality for better analysis and reporting.

The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'Schemas' tree with 'phoenixadmin' selected, containing tables like 'shipment_tbl' and 'cleaned_shipment'. The main editor window shows a SQL procedure named 'cleaned_shipment_query' with the following code:

```
1 CREATE DEFINER='root'@'localhost' PROCEDURE `cleaned_shipment_query`()
2 BEGIN
3   select shipment_id,
4     proper_case(trim(origin_warehouse)) as origin_warehouse,
5     proper_case(trim(destination_city)) as destination_city,
6     ucase(trim(destination_state)) as destination_state,
7     proper_case(carrier) as carrier,
8     proper_case(shipment_status) as shipment_status,
9     freight_cost,
10    items_count,
11    proper_case(damage_reported) as damage_reported,
12
13    str_to_date(delivery_date, '%Y-%m-%d') as delivery_date,
14    str_to_date(ship_date, '%Y-%m-%d') as ship_date,
15
16    datediff(delivery_date, ship_date) as total_days,
17
```

The bottom panel shows the 'Output' tab with a table of execution results:

#	Time	Action	Message	Duration / Fetch
64	11:22:13	Apply changes to cleaned_shipment_query	Changes applied	0.075 sec / 0.000 sec
65	11:22:34	select shipment_id, proper_case(trim(origin_warehouse)) as origin_warehouse, proper_case(trim(destination_city)) as destination_city, ucse...	25 rows returned	0.000 sec
66	11:23:59	DELETE FROM shipment_tbl where shipment_id in (select shipment_id from (select shipment_id, row_number() over (partition by origin...	Error Code: 1054. Unknown column 'row_num' in 'where clause'	0.000 sec
67	11:24:09	DELETE FROM shipment_tbl where shipment_id in (select shipment_id from (select shipment_id, row_number() over (partition by origin...	Error Code: 1054. Unknown column 'row_num' in 'where clause'	0.000 sec
68	11:24:31	DELETE FROM shipment_tbl where shipment_id in (select shipment_id from (select shipment_id, row_number() over (partition by origin...	Error Code: 1054. Unknown column 'row_num' in 'where clause'	0.000 sec
69	11:24:40	DELETE FROM shipment_tbl where shipment_id in (select shipment_id from (select shipment_id, row_number() over (partition by origin...	Error Code: 1054. Unknown column 'row_num' in 'where clause'	0.000 sec
70	11:31:29	Apply changes to cleaned_shipment	Changes applied	0.000 sec
71	11:33:20	DELETE FROM shipment_tbl where shipment_id in (select shipment_id from (select shipment_id, row_number() over (partition by origin...	Error Code: 1054. Unknown column 'row_num' in 'where clause'	0.000 sec
72	11:37:44	DELETE FROM shipment_tbl where shipment_id in (select shipment_id from (select shipment_id, row_number() over (partition by origin...	Error Code: 1054. Unknown column 'row_num' in 'where clause'	0.000 sec
73	11:37:50	Apply changes to cleaned_shipment	No changes detected	0.000 sec
74	11:37:51	Apply changes to cleaned_shipment	No changes detected	0.000 sec
75	11:38:19	Apply changes to cleaned_shipment_query	Selected name conflicts with existing procedure 'cleaned_shipment_query'.	0.000 sec
76	11:38:22	Apply changes to cleaned_shipment_query	Selected name conflicts with existing procedure 'cleaned_shipment_query'.	0.000 sec
77	11:38:36	Apply changes to cleaned_shipment_query	Selected name conflicts with existing procedure 'cleaned_shipment_query'.	0.000 sec
78	11:38:46	Apply changes to cleaned_shipment_query	Selected name conflicts with existing procedure 'cleaned_shipment_query'.	0.000 sec